

Multiple Model Architecture

This document describes how to extend the BrightSign NPU Gaze Extension architecture to support running multiple models concurrently, leveraging all 3 cores of the RK3588's NPU for retail analytics.

Executive Summary

The RK3588 NPU has 3 cores that can run models independently. To run RetinaFace (face/gaze detection), YOLOv8-pose (person pose estimation), and YOLOx (object detection) concurrently on all 3 cores, we need to:

1. **Split preprocessing into model-specific branches** after initial capture
2. **Create parallel inference pipelines** with dedicated queues and workers per model
3. **Implement NPU core affinity** to bind models to specific NPU cores
4. **Merge results in a fusion post-processor** that correlates detections
5. **Extend configuration** to support multiple ModelSpecs

The key insight: **share capture, fork preprocessing, parallelize inference, merge results.**

Retail Analytics Goals

The system provides comprehensive shopper analytics by combining outputs from three specialized models:

Detection Outputs:

- **Person count:** Number of people in frame (from YOLOv8-pose)
- **Face count:** Number of faces detected (from RetinaFace)
- **Gaze count:** Number of faces looking at screen (from RetinaFace gaze estimation)
- **Pose data:** 17 COCO keypoints per person (shoulders, elbows, wrists, hips, knees, ankles, etc.)
- **Object context:** Detected objects like carts, baskets, products (from YOLOx)

Frame-to-Frame Tracking:

- **Movement direction:** Track person position changes across frames
- **Gaze state changes:** Detect when person looks toward/away from screen
- **Pose transitions:** Identify pose changes indicating specific behaviors

Behavioral Inference (via transfer learning on YOLOv8-pose):

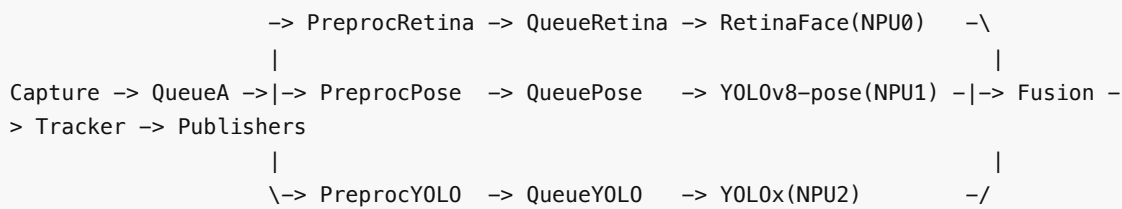
- **Pushing cart:** Arm/hand positions extended forward, body leaning slightly
- **Carrying basket:** Elbow bent, hand raised to waist/chest height
- **Shelf interaction:** Arm extension toward shelf, hand position near product level
- **Standing/browsing:** Stationary with minimal pose variation
- **Walking:** Regular leg movement pattern

Architecture Changes

Current Single-Model Architecture (Recap)

```
Capture -> QueueA -> Preprocess -> QueueB -> ModelRunner -> Postprocess -> Publishers
```

New 3-Model Architecture (Full NPU Utilization)



Key principles:

- **Single capture source:** One camera feed for all 3 models (minimizes bandwidth)
- **Forked preprocessing:** Each model gets its own preprocessing worker with model-specific transforms
- **Parallel inference:** All 3 models run concurrently, one per NPU core
- **Result fusion:** Combined post-processor merges detections with timestamp correlation
- **Temporal tracking:** Frame-to-frame tracker maintains person state, movement, and behavior history

Detailed Component Changes

1. Capture Worker (Minimal Changes)

Current behavior:

- Pulls frames from RTSP/USB
- Pushes to single FrameQueue A

New behavior:

- Pulls frames from RTSP/USB
- **Pushes to shared FrameQueue A** (no change)
- Frame is consumed by multiple preprocessing workers via **broadcast or shared pointer**

Implementation options:

Option A: Broadcast Queue

```

class BroadcastQueue {
    void push(Frame frame);
    Frame pop(int consumer_id, timeout); // Each consumer gets a copy
};
  
```

Option B: Shared Pointer Queue (Recommended)

```

// Capture pushes shared_ptr<Frame>
captureQueue.push(std::make_shared<Frame>(raw_frame));

// Multiple preproc workers pop the same shared_ptr
auto frame = captureQueue.pop(); // Reference counted, no copy
  
```

Recommendation: Use **Option B** (shared pointers) to minimize memory copies. The frame buffer is immutable after capture, so multiple readers are safe.

2. Preprocessing Pipeline (Fork into Model-Specific Workers)

Current behavior:

- Single preprocessing worker
- Reads ModelSpec to determine transforms
- Converts NV12->RGB, resizes, normalizes

New behavior:

- **Multiple preprocessing workers**, one per model
- Each worker reads its own ModelSpec configuration
- Workers run in parallel threads

Example: Three preprocessing workers

```
// RetinaFace preprocessing worker
class RetinaPreprocessWorker {
    ModelSpec retinaSpec; // 320x320 RGB, mean=[104,117,123]

    void run() {
        while (running) {
            auto frame = captureQueue.pop(); // shared_ptr<Frame>
            auto tensor = preprocess(frame, retinaSpec);
            retinaQueue.push(tensor);
        }
    }
};

// YOLOv8-pose preprocessing worker
class PosePreprocessWorker {
    ModelSpec poseSpec; // 640x640 BGR, mean=[0,0,0], std=[255,255,255]

    void run() {
        while (running) {
            auto frame = captureQueue.pop(); // same shared_ptr<Frame>
            auto tensor = preprocess(frame, poseSpec);
            poseQueue.push(tensor);
        }
    }
};

// YOLOx preprocessing worker
class YOLOPreprocessWorker {
    ModelSpec yoloSpec; // 640x640 BGR, mean=[0,0,0], std=[255,255,255]

    void run() {
        while (running) {
            auto frame = captureQueue.pop(); // same shared_ptr<Frame>
            auto tensor = preprocess(frame, yoloSpec);
            yoloQueue.push(tensor);
        }
    }
};
```

Key considerations:

- Each worker maintains its own RGA context for parallel hardware acceleration
- Workers are independent threads; no synchronization needed (they consume the same input)
- Original frame from CaptureQueue is **immutable** (only read, never modified)

3. Inference Queues (One per Model)

Current behavior:

- Single FrameQueue B (capacity=1, overwrite policy)

New behavior:

- **Multiple inference queues**, one per model
- Each queue has capacity=1 with overwrite/drop-old policy

```
FrameQueue retinaInferenceQueue(capacity=1, policy=OVERWRITE);
FrameQueue poseInferenceQueue(capacity=1, policy=OVERWRITE);
FrameQueue yoloInferenceQueue(capacity=1, policy=OVERWRITE);
```

Why separate queues?

- Models have different inference times (RetinaFace ~15ms, YOLOv8-pose ~25ms, YOLOx ~30ms)
- Independent queues prevent head-of-line blocking
- Each model gets freshest preprocessed frame at its own pace

4. ModelRunner (Parallel Instances with NPU Core Affinity)

Current behavior:

- Single ModelRunner with IModel interface
- RKNN2Interface manages NPU device/session

New behavior:

- **Multiple ModelRunner instances**, one per model
- Each runner explicitly binds to a specific NPU core

NPU Core Affinity Implementation:

```
class ModelRunner {
    IModel* model;
    int npu_core_id; // 0, 1, or 2 for RK3588

    void initialize() {
        // RKNN allows specifying target core during init
        rknn_init_opt opt;
        opt.target_core = npu_core_id; // Bind to specific NPU core

        rknn_init(&context, model_data, model_size, 0, &opt);
    }

    void run() {
        while (running) {
```

```
        auto tensor = inferenceQueue.pop();
        auto result = model->infer(tensor);
        postprocessQueue.push(result);
    }
}

};

// Usage
ModelRunner retinaRunner(retinaModel, npu_core_id=0);
ModelRunner poseRunner(poseModel, npu_core_id=1);
ModelRunner yoloRunner(yoloModel, npu_core_id=2);
```

RK3588 NPU Core Allocation Strategy:

NPU Core	Assignment	Rationale
Core 0	RetinaFace	Face/gaze detection (fastest, ~15ms)
Core 1	YOLOv8-pose	Pose estimation with 17 keypoints (~25ms)
Core 2	YOLOx	Object detection for carts/baskets/products (~30ms)

Important: RKNN API allows core affinity via `rknn_init` flags. Verify with RKNN toolkit that models are actually scheduled on target cores (check `rknn_query(RKNN_QUERY_PERF_DETAIL)`).

5. Post-Processing (Fusion of Multi-Model Results)

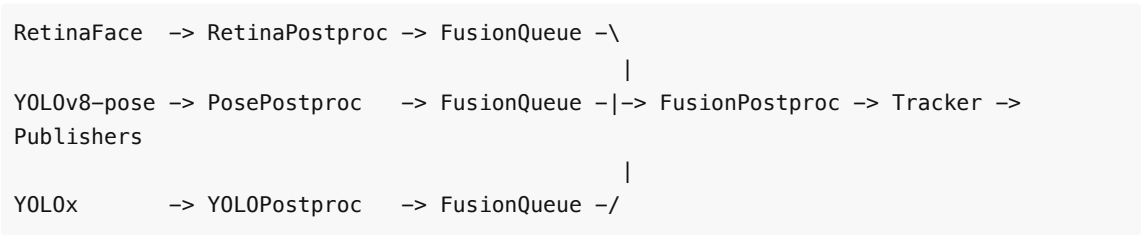
Current behavior:

- Single post-processor for one model's output
- Extracts faces/gaze, applies NMS

New behavior:

- **Parallel post-processors** per model (initial processing)
- **Fusion post-processor** that merges results

Architecture:



Implementation:

```
// Per-model post-processors (existing logic)
class RetinaPostprocessor {
    void process(RetinaOutput output) {
        auto faces = extractFaces(output); // NMS, thresholds
        auto gazes = computeGaze(faces);
```

```

        DetectionResult result;
        result.timestamp = output.timestamp;
        result.faces = faces;
        result.gazes = gazes;
        result.gaze_count = countLookingAtScreen(gazes);

        fusionQueue.push(result);
    }
};

class PosePostprocessor {
    void process(PoseOutput output) {
        auto persons = extractPersons(output); // NMS, confidence threshold

        // Extract 17 COCO keypoints per person
        // [nose, left_eye, right_eye, left_ear, right_ear,
        //  left_shoulder, right_shoulder, left_elbow, right_elbow,
        //  left_wrist, right_wrist, left_hip, right_hip,
        //  left_knee, right_knee, left_ankle, right_ankle]

        DetectionResult result;
        result.timestamp = output.timestamp;
        result.persons = persons;
        result.person_count = persons.size();

        fusionQueue.push(result);
    }
};

class YOLOPostprocessor {
    void process(YOLOOutput output) {
        auto objects = extractObjects(output); // NMS, class filtering

        DetectionResult result;
        result.timestamp = output.timestamp;
        result.objects = objects; // carts, baskets, products

        fusionQueue.push(result);
    }
};

// Fusion post-processor (new component)
class FusionPostprocessor {
    struct TimestampedResult {
        uint64_t timestamp;
        optional<FaceDetections> faces;
        optional<PoseDetections> persons;
        optional<ObjectDetections> objects;

        int person_count = 0;
        int face_count = 0;
        int gaze_count = 0;
    };
};

```

```

};

// Buffer to hold results waiting for pairing
std::map<uint64_t, TimestampedResult> buffer;

void process() {
    while (running) {
        auto result = fusionQueue.pop();

        // Find or create entry for this timestamp
        auto& entry = buffer[result.timestamp];

        // Merge into entry
        if (result.has_faces()) {
            entry.faces = result.faces;
            entry.face_count = result.faces.size();
            entry.gaze_count = result.gaze_count;
        }
        if (result.has_persons()) {
            entry.persons = result.persons;
            entry.person_count = result.person_count;
        }
        if (result.has_objects()) {
            entry.objects = result.objects;
        }

        // Check if we have all expected results for this timestamp
        if (entry.is_complete() || entry.is_too_old()) {
            // Correlate detections
            correlatePersonsWithFaces(entry);
            correlatePersonsWithObjects(entry);

            // Send to tracker
            trackerQueue.push(entry);
            buffer.erase(result.timestamp);
        }
    }
}

void correlatePersonsWithFaces(TimestampedResult& entry) {
    // Match face bboxes with person bboxes based on IoU
    // Link gaze direction to specific person
}

void correlatePersonsWithObjects(TimestampedResult& entry) {
    // Associate nearby carts/baskets with persons
    // Detect if person's hand keypoints overlap with shelf products
}
};

```

Fusion strategies:

1. **Timestamp matching** (primary): Wait for results from all 3 models with matching timestamps
2. **Timeout-based**: If one model is slow, publish partial results after timeout (e.g., 100ms)
3. **Best-effort**: Publish latest available results from each model, even if timestamps don't align

Correlation logic:

- **Associate faces with persons**: Match face bbox with person bbox using IoU, link gaze to specific person
- **Associate persons with objects**: Link nearby carts/baskets to persons based on proximity
- **Detect shelf interactions**: Check if wrist keypoints are near product bounding boxes
- **Scene understanding**: Combine all detections for rich behavioral context

6. Temporal Tracking (New Component)

The tracker maintains frame-to-frame continuity and detects behavioral patterns over time.

Implementation:

```
class TemporalTracker {
    struct TrackedPerson {
        int track_id;
        cv::Rect2f bbox;
        std::vector<cv::Point2f> keypoints; // 17 COCO keypoints

        // Face/gaze association
        optional<int> face_id;
        bool looking_at_screen;

        // Object associations
        optional<int> cart_id;
        optional<int> basket_id;

        // Movement tracking
        cv::Point2f velocity; // pixels/frame
        float direction_angle; // radians

        // Behavior state
        PoseBehavior behavior; // PUSHING_CART, CARRYING_BASKET, SHELF_INTERACTION,
etc.
        int behavior_frame_count; // frames in current behavior

        // History for temporal analysis
        std::deque<cv::Point2f> position_history; // last N positions
        std::deque<PoseKeypoints> pose_history; // last N poses
        int frames_since_gaze_change;
    };

    std::map<int, TrackedPerson> active_tracks;
    int next_track_id = 0;

    void update(const TimestampedResult& fused_result) {
        // 1. Data association: match current detections to existing tracks
        auto associations = matchDetectionsToTracks(fused_result.persons);
```



```

// 2. Update existing tracks
for (auto& [det_idx, track_id] : associations) {
    updateTrack(track_id, fused_result, det_idx);
}

// 3. Create new tracks for unmatched detections
for (auto& unmatched_det : getUnmatchedDetections(associations)) {
    createNewTrack(unmatched_det, fused_result);
}

// 4. Remove stale tracks (not detected for N frames)
removeStale Tracks();

// 5. Analyze behavior changes
for (auto& [track_id, person] : active_tracks) {
    analyzeBehaviorChange(person);
    detectMovementPattern(person);
    trackGazeChanges(person);
}

// 6. Publish tracking results
publishTrackingResults();
}

void updateTrack(int track_id, const TimestampedResult& result, int det_idx) {
    auto& track = active_tracks[track_id];
    auto& detection = result.persons[det_idx];

    // Update position
    cv::Point2f prev_center = getCenter(track.bbox);
    track.bbox = detection.bbox;
    cv::Point2f new_center = getCenter(detection.bbox);

    // Compute velocity
    track.velocity = new_center - prev_center;
    track.direction_angle = atan2(track.velocity.y, track.velocity.x);

    // Update keypoints and pose history
    track.keypoints = detection.keypoints;
    track.pose_history.push_back(detection.keypoints);
    if (track.pose_history.size() > 10) track.pose_history.pop_front();

    // Update position history
    track.position_history.push_back(new_center);
    if (track.position_history.size() > 30) track.position_history.pop_front();

    // Associate with face/gaze
    track.face_id = findMatchingFace(track.bbox, result.faces);
    if (track.face_id) {
        bool prev_gaze = track.looking_at_screen;
        track.looking_at_screen =

```

```

result.faces[*track.face_id].looking_at_screen;
    if (prev_gaze != track.looking_at_screen) {
        track.frames_since_gaze_change = 0;
    } else {
        track.frames_since_gaze_change++;
    }
}

// Associate with objects (cart, basket)
track.cart_id = findNearbyObject(track.bbox, result.objects, "cart");
track.basket_id = findNearbyObject(track.bbox, result.objects, "basket");
}

void analyzeBehaviorChange(TrackedPerson& person) {
    // Infer behavior from pose keypoints and object associations
    PoseBehavior new_behavior = inferBehavior(person);

    if (new_behavior != person.behavior) {
        // Behavior changed - emit event
        emitBehaviorChangeEvent(person.track_id, person.behavior, new_behavior);
        person.behavior = new_behavior;
        person.behavior_frame_count = 0;
    } else {
        person.behavior_frame_count++;
    }
}

PoseBehavior inferBehavior(const TrackedPerson& person) {
    // Use heuristics on keypoints + object associations
    // Later: use transfer-learned classifier

    if (person.cart_id.has_value()) {
        // Check arm positions for pushing gesture
        if (armsExtendedForward(person.keypoints)) {
            return PoseBehavior::PUSHING_CART;
        }
    }

    if (person.basket_id.has_value()) {
        // Check for bent elbow, raised hand
        if (armBentWithRaisedHand(person.keypoints)) {
            return PoseBehavior::CARRYING_BASKET;
        }
    }

    // Check for shelf interaction (hand near products)
    if (handNearShelfProducts(person, objects)) {
        return PoseBehavior::SHELF_INTERACTION;
    }

    // Check movement
    if (person.velocity.norm() < 2.0) {

```

```

        return PoseBehavior::STANDING;
    } else {
        return PoseBehavior::WALKING;
    }
}
};

```

Tracking Features:

- **Person re-identification:** Maintains consistent track IDs across frames using IoU + keypoint similarity
- **Velocity estimation:** Computes direction and speed of movement
- **Gaze tracking:** Monitors when person starts/stops looking at screen
- **Behavior classification:** Infers activity from pose + object context
- **Event detection:** Emits events for behavior changes (started browsing, picked up product, etc.)

Transfer Learning Note:

The `inferBehavior()` function initially uses heuristics. For production, train a classifier on the 17 keypoints to recognize:

- Pushing cart (arms extended, torso leaning)
- Carrying basket (elbow bent, hand raised)
- Shelf reach (arm extension, hand up)
- Product pickup (hand near shelf, then retraction)

Fine-tune YOLOv8-pose or add a lightweight MLP head on the keypoints for behavior classification.

7. Resource Manager (Extended for Multi-Model)

Current behavior:

- Manages RGA contexts, tensor buffers for single model

New behavior:

- Manages resources for **multiple models**
- Ensures NPU cores don't exceed capacity
- Pools tensors per model

Changes:

```

class ResourceManager {
    // Per-model RGA contexts
    std::map<std::string, rga_context> rgaContexts;

    // Per-model tensor pools
    std::map<std::string, TensorPool> tensorPools;

    // NPU core allocation tracker
    std::bitset<3> npuCoresInUse; // RK3588 has 3 cores

    int allocateNPUCore() {
        for (int i = 0; i < 3; i++) {
            if (!npuCoresInUse[i]) {

```

```

        npuCoresInUse.set(i);
        return i;
    }
}
throw std::runtime_error("No NPU cores available");
}

void releaseNPUCore(int core_id) {
    npuCoresInUse.reset(core_id);
}
};

```

Memory considerations:

- Each model needs separate input/output tensors
- RetinaFace: ~300KB input, ~50KB output
- YOLOx: ~4.7MB input (640x640x3 float), ~200KB output
- Total NPU memory budget on RK3588: ~800MB shared across cores

8. Configuration (Multi-Model Support)

Current configuration:

```

processing:
  model: "retinaface_rknn"
  preprocessing: { ... }

```

New multi-model configuration:

```

processing:
  # Enable multi-model mode
  multi_model: true

models:
  - name: "retinaface"
    enabled: true
    model_path: "/models/retinaface_320.rknn"
    npu_core: 0 # Explicit core assignment
    priority: "high" # For scheduling hints

    preprocessing:
      target_size: [320, 320]
      color_format: "RGB"
      normalize:
        mean: [104, 117, 123]
        std: [1, 1, 1]

    postprocessing:
      type: "retinaface"
      nms_threshold: 0.4
      confidence_threshold: 0.7
      max_faces: 10

```

```

- name: "yolov8_pose"
  enabled: true
  model_path: "/models/yolov8n_pose_640.rknn"
  npu_core: 1 # Core 1 for pose estimation
  priority: "high"

  preprocessing:
    target_size: [640, 640]
    color_format: "BGR"
    normalize:
      mean: [0, 0, 0]
      std: [255, 255, 255]
    letterbox: true

  postprocessing:
    type: "yolov8_pose"
    nms_threshold: 0.5
    confidence_threshold: 0.5
    max_persons: 20
    keypoint_confidence_threshold: 0.3

- name: "yolox"
  enabled: true
  model_path: "/models/yolox_s_640.rknn"
  npu_core: 2 # Core 2 for object detection
  priority: "medium"

  preprocessing:
    target_size: [640, 640]
    color_format: "BGR"
    normalize:
      mean: [0, 0, 0]
      std: [255, 255, 255]
    letterbox: true

  postprocessing:
    type: "yolox"
    nms_threshold: 0.5
    confidence_threshold: 0.5
    classes: ["cart", "basket", "bottle", "cup", "cell phone"] # Retail-
relevant objects

# Fusion configuration
fusion:
  enabled: true
  timestamp_tolerance_ms: 50 # Max time diff for matching results
  timeout_ms: 100 # Max wait for missing results
  correlate_persons_faces: true # Link persons with face detections
  correlate_persons_objects: true # Link persons with carts/baskets
  detect_shelf_interactions: true # Check wrist keypoints near products

```

```

# Tracking configuration
tracking:
  enabled: true
  max_age: 30 # frames to keep track without detection
  min_hits: 3 # detections required before publishing track
  iou_threshold: 0.3 # for matching detections to tracks

# Behavior classification
behavior_detection:
  enabled: true
  use_transfer_learning: false # Set true when model is trained
  model_path: "/models/behavior_classifier.rknn" # Optional: trained classifier

output:
  publishers:
    - type: "udp_json"
      port: 8080
      include_faces: true
      include_persons: true
      include_person_count: true
      include_face_count: true
      include_gaze_count: true
      include_objects: true
      include_tracks: true
      include_behaviors: true
      include_correlations: true

```

9. Orchestrator (Multi-Pipeline Management)

Current behavior:

- Starts single capture -> preprocess -> infer -> postprocess pipeline

New behavior:

- Starts capture (single)
- **Starts 3 preprocessing workers** (one per model)
- **Starts 3 inference workers** (one per model, one per NPU core)
- **Starts 3 postprocessing workers** (one per model)
- Starts fusion worker
- Starts tracking worker
- Starts publishers

Thread topology:

```

Thread 1: CaptureWorker
Thread 2: RetinaPreprocessWorker
Thread 3: RetinaInferenceWorker (NPU core 0)
Thread 4: RetinaPostprocessWorker
Thread 5: PosePreprocessWorker
Thread 6: PoseInferenceWorker (NPU core 1)
Thread 7: PosePostprocessWorker
Thread 8: YOLOPreprocessWorker

```

```
Thread 9: YOLOInferenceWorker (NPU core 2)
Thread 10: YOLOPostprocessWorker
Thread 11: FusionWorker
Thread 12: TrackingWorker
Thread 13: PublisherWorker
```

Total: 13 threads (vs. 4-5 for single model)

Orchestrator initialization:

```
class Orchestrator {
    void start() {
        // Start capture (shared by all models)
        captureWorker = std::make_unique<CaptureWorker>(config.input);
        captureWorker->start();

        // Start per-model pipelines
        for (auto& modelConfig : config.models) {
            if (!modelConfig.enabled) continue;

            // Allocate NPU core
            int core = resourceMgr.allocateNPUCore();

            // Create preprocessing worker
            auto preproc = std::make_unique<PreprocessWorker>(
                captureQueue, modelConfig.preprocessing
            );
            preprocWorkers.push_back(std::move(preproc));

            // Create inference worker
            auto infer = std::make_unique<InferenceWorker>(
                modelConfig.model_path, core, modelConfig.preprocessing
            );
            inferWorkers.push_back(std::move(infer));

            // Create postprocessing worker
            auto postproc = std::make_unique<PostprocessWorker>(
                modelConfig.postprocessing
            );
            postprocWorkers.push_back(std::move(postproc));
        }

        // Start fusion worker
        if (config.fusion.enabled) {
            fusionWorker = std::make_unique<FusionWorker>(config.fusion);
            fusionWorker->start();
        }

        // Start tracking worker
        if (config.tracking.enabled) {
            trackingWorker = std::make_unique<TrackingWorker>(config.tracking);
            trackingWorker->start();
        }
    }
};
```

```

    }

    // Start publishers
    for (auto& pub : publishers) {
        pub->start();
    }
}
};

```

Data Flow Diagram

Multi-Model Steady State Processing

```

sequenceDiagram
    autonumber
    participant Cap as "Capture"
    participant QA as "CaptureQueue"
    participant PR as "PreprocRetina"
    participant PP as "PreprocPose"
    participant PY as "PreprocYOLO"
    participant QR as "QueueRetina"
    participant QP as "QueuePose"
    participant QY as "QueueYOLO"
    participant MR as "RetinaFace(NPU0)"
    participant MP as "YOLOv8-pose(NPU1)"
    participant MY as "YOLOx(NPU2)"
    participant PostR as "PostRetina"
    participant PostP as "PostPose"
    participant PostY as "PostYOLO"
    participant Fuse as "Fusion"
    participant Track as "Tracker"
    participant Pub as "Publishers"

    Cap->>QA: push shared_ptr<Frame>

    par Parallel Preprocessing (3 workers)
        PR->>QA: pop shared_ptr<Frame>
        PR->>PR: NV12->RGB, resize 320x320
        PR->>QR: push tensor
    and
        PP->>QA: pop shared_ptr<Frame>
        PP->>PP: NV12->BGR, resize 640x640
        PP->>QP: push tensor
    and
        PY->>QA: pop shared_ptr<Frame>
        PY->>PY: NV12->BGR, resize 640x640
        PY->>QY: push tensor
    end

    par Parallel Inference (3 NPU cores)

```



```

MR-->QR: pop tensor
MR-->MR: infer on NPU core 0
MR-->PostR: send detections
and
MP-->QP: pop tensor
MP-->MP: infer on NPU core 1
MP-->PostP: send detections
and
MY-->QY: pop tensor
MY-->MY: infer on NPU core 2
MY-->PostY: send detections
end

par Parallel Post-processing (3 workers)
PostR-->PostR: extract faces/gaze, NMS
PostR-->Fuse: push {ts, faces, gaze_count}
and
PostP-->PostP: extract persons/keypoints, NMS
PostP-->Fuse: push {ts, persons, person_count}
and
PostY-->PostY: extract objects, NMS
PostY-->Fuse: push {ts, objects}
end

Fuse-->Fuse: match by timestamp
Fuse-->Fuse: correlate persons->faces->objects
Fuse-->Track: send fused result

Track-->Track: update tracks, compute velocity
Track-->Track: infer behaviors, detect events
Track-->Pub: publish tracks + analytics

```

Performance Considerations

Throughput Analysis

Single-model baseline (RetinaFace only):

- Capture: 33ms (30 FPS)
- Preprocess: 5ms
- Inference: 15ms
- Postprocess: 2ms
- **Total latency:** ~55ms
- **Throughput:** ~18 FPS (limited by capture)

3-model (RetinaFace + YOLOv8-pose + YOLOx):

- Capture: 33ms (30 FPS) **shared**
- Preprocess: 5ms (Retina) + 8ms (Pose) + 8ms (YOLO) **parallel**
- Inference: 15ms (Retina, NPU0) + 25ms (Pose, NPU1) + 30ms (YOLO, NPU2) **parallel**
- Postprocess: 2ms (Retina) + 3ms (Pose) + 3ms (YOLO) **parallel**
- Fusion: 1ms

- Tracking: 2ms
- **Total latency:** 33ms (capture) + max(8ms preproc, 30ms infer, 3ms postproc) + 1ms fusion + 2ms tracking = **69ms**
- **Throughput:** ~14 FPS (limited by YOLOx inference on NPU core 2)

Bottleneck: YOLOx inference at 30ms on NPU core 2. Options:

- Use smaller YOLOx variant (YOLOx-nano: ~10ms) → achieve ~20 FPS
- Reduce YOLOx input size (640→416) → ~20ms inference
- Run YOLOx at reduced frame rate (every 2nd frame) while maintaining pose/face at full rate

All 3 NPU cores fully utilized

Memory Footprint

Component	Single Model	3-Model System	Delta
Model weights (RKNN)	1.5 MB (Retina)	1.5 MB + 3.5 MB (Pose) + 9 MB (YOLOx)	+12.5 MB
Input tensors	0.3 MB	0.3 MB + 4.7 MB (Pose) + 4.7 MB (YOLO)	+9.4 MB
Output tensors	0.05 MB	0.05 MB + 0.3 MB (Pose) + 0.2 MB (YOLO)	+0.5 MB
Frame buffers (shared)	1.2 MB	1.2 MB	0
Tracking state	0	~0.5 MB (30 tracks × ~17KB each)	+0.5 MB
Total	~3 MB	~26 MB	+23 MB

RK3588 NPU memory budget: ~800 MB total, so 26 MB is well within limits.

CPU Load

- Preprocessing is CPU/RGA-bound (if RGA is saturated, will use OpenCV fallback)
- With 3 preprocessing workers, RGA utilization is higher
- Tracking adds CPU overhead (~2ms per frame for 10-20 tracked persons)
- Monitor RGA queue depth; if high, consider staggering preprocessing or reducing resolution

Synchronization Overhead

- **Minimal:** Queues use lock-free primitives (if implemented correctly)
- **Fusion buffer:** Simple map lookup by timestamp, O(1) amortized
- **Shared pointer refcounting:** Atomic increment/decrement, negligible overhead

Alternative Architectures

Option A: Sequential Models (Pipeline Chaining)

Capture -> Preproc -> RetinaFace(NPU0) -> Preproc -> YOLOx(NPU1) -> Postprocess

Pros: Simpler code, no fusion needed **Cons:** Higher latency (models run sequentially), wastes NPU cores

Not recommended for real-time applications.

Option B: Time-Sliced Models (Alternating Frames)

```
Frame 0: Capture -> RetinaFace(NPU0)
Frame 1: Capture -> YOLOx(NPU0)
Frame 2: Capture -> RetinaFace(NPU0)
...
```

Pros: Uses single NPU core, simpler resource management **Cons:** Effective frame rate per model halved (15 FPS each), doesn't leverage 3-core NPU

Not recommended when multi-core NPU is available.

Option C: Hierarchical Models (Detector + Refinement)

```
Capture -> YOLOx(NPU0, detect persons) -> ROI crop -> RetinaFace(NPU1, detect faces in ROI)
```

Pros: Reduces RetinaFace search space, more efficient **Cons:** RetinaFace depends on YOLOx, introduces latency, complex ROI management

Consider for future optimization if full-frame face detection is too expensive.

Implementation Checklist

- ☐ Modify CaptureWorker to push `shared_ptr<Frame>` instead of raw frames
- ☐ Implement multiple PreprocessWorker instances with model-specific configs
- ☐ Create per-model inference queues
- ☐ Extend ModelRunner to accept NPU core affinity parameter
- ☐ Implement FusionPostprocessor with timestamp-based correlation
- ☐ Add multi-model configuration schema to YAML parser
- ☐ Update ResourceManager to track NPU core allocation
- ☐ Modify Orchestrator to spawn multiple model pipelines
- ☐ Extend metrics to track per-model FPS and latency
- ☐ Add health monitoring for per-model failures
- ☐ Implement dynamic model enable/disable (optional)
- ☐ Test NPU core affinity with RKNN profiling tools
- ☐ Validate timestamp synchronization across models
- ☐ Benchmark memory usage and CPU/NPU utilization

Testing Strategy

1. Unit tests:

- FusionPostprocessor timestamp matching logic
- ResourceManager NPU core allocation
- Configuration parsing for multi-model specs

2. Integration tests:

- Run RetinaFace + YOLOx concurrently, verify outputs
- Test with mismatched frame rates (e.g., YOLOx slower than Retina)
- Verify NPU core isolation (use RKNN perf query)

3. Performance tests:

- Measure end-to-end latency with 1 model vs. 2 models
- Profile CPU usage per worker thread
- Measure NPU utilization per core
- Test memory stability over 24-hour run

4. Failure tests:

- One model crashes, verify other model continues
- NPU core hangs, verify recovery mechanism
- One model falls behind, verify fusion timeout behavior

Future Extensions

3-Model Configuration

Add a third model (e.g., pose estimation) using NPU core 2:

```
models:
- name: "retinaface"
  npu_core: 0
- name: "yolox"
  npu_core: 1
- name: "movenet" # Pose estimation
  npu_core: 2
```

Dynamic Model Loading

Support runtime enable/disable of models without restarting pipeline:

```
orchestrator.enableModel("yolox");
orchestrator.disableModel("movenet");
```

Adaptive Frame Rate

Reduce slower model's frame rate to maintain overall throughput:

```
models:
- name: "yolox"
  frame_decimation: 2 # Run on every 2nd frame
```

Cross-Model Dependencies

Allow one model's output to influence another's processing:

```
models:
- name: "yolox"
  output_to: ["retinaface"] # YOLO person detections -> RetinaFace ROIs
```

Conclusion

Running multiple models on the RK3588's 3-core NPU requires:

1. **Shared capture** with immutable frame buffers (shared pointers)
2. **Forked preprocessing** with model-specific workers
3. **Parallel inference** with explicit NPU core affinity
4. **Timestamp-based fusion** to correlate results
5. **Extended configuration** for multi-model specs

The architecture maintains the original design principles (modularity, real-time performance, reliability) while scaling to multiple concurrent models. The fusion layer adds minimal latency (<1ms) and enables rich multi-modal outputs (faces + objects + correlations).

Key insight: The existing pipeline architecture is **already well-suited** for multi-model extension. The IModel interface, worker-based threading, and queue-based communication make it straightforward to fork the pipeline after capture and merge results before publishing.